

# Dataset expansion: benchmarking candidate HF corpora

Question: which HuggingFace datasets raise the difficulty ceiling of the LaTeX->Typst benchmark?

Probe: 1-turn sonnet-4.6 (low effort) on every set — same prompt & official scoring as harness\_baseline — then an agentic check (opus low, visual, harness v3, \$3 cap) on 6 hard picks (short + verbose).

7 new sets built from: stanford-crfm/image2struct-latex-v1 (equation/table/algorithm/plot), deepcopy/pubmed-tables-latex-768px, TIGER-Lab/arxiv-latex-5T, Mithilss/neurips-2025-arxiv-latex-sources.

Caveat: new reference PDFs compiled with tectonic 0.16.9 (XeTeX), not pdfLaTeX — small font-metric drift vs the lathe canon; acceptable for difficulty ranking.

# 1-turn difficulty probe

sonnet-4.6, low effort, ~\$0.06-0.43/run

Compile rate is the primary signal (a set nobody compiles is past the current ceiling);  
component means are over compiled samples only.

| set           | n  | compiled | overall | content | layout | raster | note                            |
|---------------|----|----------|---------|---------|--------|--------|---------------------------------|
| lathe_overall | 11 | 6/11     | 49.7    | 89.7    | 48.9   | 13.2   | current benchmark, 1/category   |
| lathe_hard    | 8  | 6/8      | 44.0    | 88.4    | 38.1   | 10.2   | current hard set (core 6+2)     |
| i2s_equation  | 5  | 1/5      | 57.7    | 64.4    | 68.0   | 24.7   | image2struct equations          |
| i2s_table     | 5  | 3/5      | 68.1    | 92.9    | 72.8   | 27.2   | image2struct tables             |
| i2s_algorithm | 5  | 5/5      | 65.3    | 87.4    | 73.6   | 24.8   | image2struct algorithms         |
| i2s_plot      | 5  | 0/5      | —       | —       | —      | —      | TikZ/pgfplots — new category    |
| pubmed_table  | 5  | 2/5      | 29.8    | 81.1    | 28.3   | 3.2    | PubMed clinical tables          |
| arxiv5t_paper | 5  | 1/5      | 4.1     | 0.9     | 23.7   | 0.4    | arXiv full papers, real figures |
| neurips_paper | 2  | 0/2      | —       | —       | —      | —      | NeurIPS full papers             |

Ranking: full papers (1/7 compile; the one compile scored 4.1) > i2s\_plot (0/5) > pubmed\_table (2/5, 29.8)  
> i2s\_equation (1/5) >> i2s\_table / i2s\_algorithm (near or easier than the current hard set).

# Agentic check: does the harness solve them?

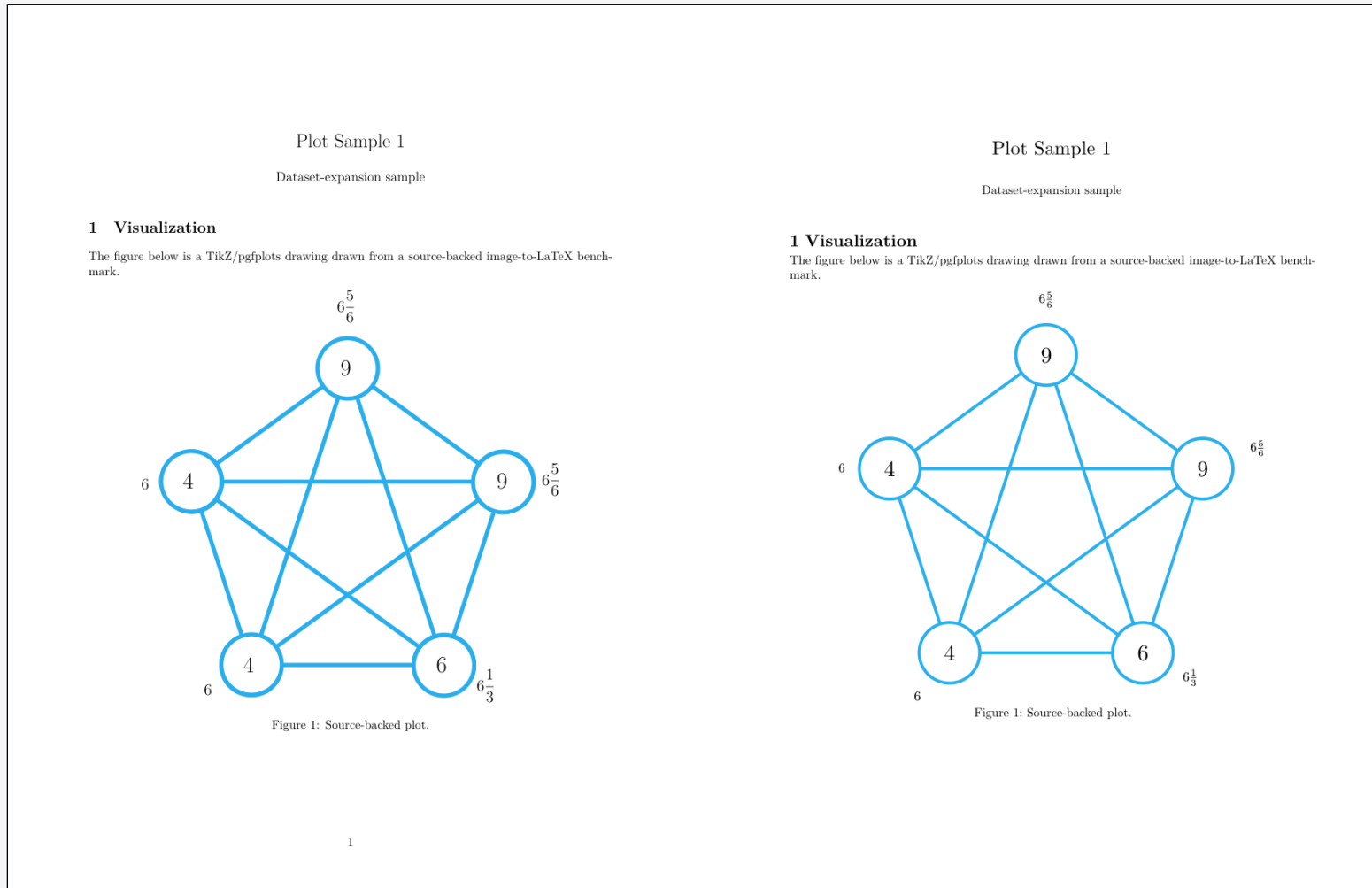
opus low, visual, harness v3, \$3 cap

6 hard picks (short + verbose). 1-turn compiled 1/6; the harness compiled 6/6 — but plateaued at 57.9-67.1, all below the ~78 the same harness family reaches on the lathe hard set. Every run stopped on plateau, not budget.

| sample            | 1-turn | overall | content | layout | raster | pages | cost   | time  | turns |
|-------------------|--------|---------|---------|--------|--------|-------|--------|-------|-------|
| arxiv5t_paper_019 | nc     | 57.9    | 94.5    | 63.7   | 14.3   | 5/4   | \$0.70 | 2.6m  | 13    |
| i2s_equation_001  | nc     | 58.6    | 60.7    | 82.6   | 22.6   | 1/1   | \$0.78 | 2.5m  | 23    |
| i2s_plot_001      | nc     | 67.1    | 98.4    | 91.3   | 16.6   | 1/1   | \$0.80 | 2.9m  | 20    |
| neurips_paper_029 | nc     | 62.4    | 83.8    | 51.9   | 31.6   | 11/11 | \$2.16 | 12.3m | 24    |
| pubmed_table_004  | 27.3   | 60.2    | 77.8    | 53.4   | 33.5   | 2/2   | \$0.82 | 2.6m  | 22    |
| pubmed_table_005  | nc     | 58.0    | 88.6    | 57.9   | 18.7   | 2/2   | \$0.87 | 2.9m  | 22    |

Two hardness flavors emerge:

verbose-hard (papers): content fine, style/pagination breaks; short-hard (equations): content itself stalls.



reference (left) vs harness candidate (right)

1-turn: nothing in this set compiles — TikZ has no direct Typst analog.

Harness rebuilds the graph with native Typst curve/circle primitives:

nodes, edges, colors, labels all present (content 98.4).

Residual divergence: figure drawn slightly smaller and shifted, edge-label

fractions collapse, node-label anchors drift, heading typography differs.

Hardness = precise coordinate-space fidelity, not TikZ translation per se.

Equation Sample 1

Dataset-expansion sample

## 1 Derivation

The following display is drawn from a source-backed image-to-Latex benchmark and reproduced verbatim.

$$\begin{aligned}
 \alpha_3 &\stackrel{(i)}{\leq} \sum_{s \in \mathcal{S}, a \in \mathcal{A}} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \sqrt{\frac{C_{\text{clipped}}^* \log \frac{N}{\delta}}{N \min \left\{ d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho), \frac{1}{S(A+B)} \right\}}} \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-) \\
 &\leq \sqrt{\frac{C_{\text{clipped}}^* \log \frac{N}{\delta}}{N}} \sum_{s \in \mathcal{S}, a \in \mathcal{A}} \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\
 &\quad + \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sum_{s \in \mathcal{S}, a \in \mathcal{A}} \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \cdot \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)}} \\
 &\stackrel{(ii)}{\leq} \sqrt{\frac{C_{\text{clipped}}^* \log \frac{N}{\delta}}{N}} \cdot \sqrt{SA} \cdot \sqrt{\sum_{s \in \mathcal{S}, a \in \mathcal{A}} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\
 &\quad + \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \left[ \sum_{s \in \mathcal{S}, a \in \mathcal{A}} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \right] \sqrt{\sum_{s \in \mathcal{S}, a \in \mathcal{A}} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\
 &\leq 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sqrt{\sum_{s \in \mathcal{S}, a \in \mathcal{A}} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\
 &\stackrel{(iii)}{=} 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sqrt{\sum_{s \in \mathcal{S}} d^{\mu^*, \nu_0}(s; \rho) \mathbb{E}_{a \sim \mu^*(s), b \sim \nu_0(s)} [\text{Var}_{P_{s,a,b}}(V_{\text{pe}}^-)]} \\
 &\stackrel{(iv)}{\leq} 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sqrt{\sum_{s \in \mathcal{S}} d^{\mu^*, \nu_0}(s; \rho) \text{Var}_{P_{\mu^*, \nu_0}}(V_{\text{pe}}^-)}.
 \end{aligned}$$

1

Equation Sample 1

Dataset-expansion sample

## Derivation

The following display is drawn from a source-backed image-to-Latex benchmark and reproduced verbatim.

$$\begin{aligned}
 \alpha_3 &\stackrel{(i)}{\leq} \sum_{s \in \mathcal{S}, a \in \mathcal{A}} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \sqrt{\frac{C_{\text{clipped}}^* \log \frac{N}{\delta}}{N \min \left\{ d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho), \frac{1}{S(A+B)} \right\}}} \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-) \\
 &\leq \sqrt{\frac{C_{\text{clipped}}^* \log \frac{N}{\delta}}{N}} \sum_{s \in \mathcal{S}, a \in \mathcal{A}} \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\
 &\quad + \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sum_{s \in \mathcal{S}, a \in \mathcal{A}} \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \cdot \sqrt{d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)}} \\
 &\stackrel{(ii)}{\leq} \sqrt{\frac{C_{\text{clipped}}^* \log \frac{N}{\delta}}{N}} \cdot \sqrt{SA} \cdot \sqrt{\sum_{s \in \mathcal{S}, a \in \mathcal{A}} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\
 &\quad + \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \left[ \sum_{s \in \mathcal{S}, a \in \mathcal{A}} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \right] \sqrt{\sum_{s \in \mathcal{S}, a \in \mathcal{A}} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\
 &\leq 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sqrt{\sum_{s \in \mathcal{S}, a \in \mathcal{A}} d^{\mu^*, \nu_0}(s, a, \nu_0(s); \rho) \text{Var}_{P_{s,a,\nu_0(s)}}(V_{\text{pe}}^-)} \\
 &\stackrel{(iii)}{=} 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sqrt{\sum_{s \in \mathcal{S}} d^{\mu^*, \nu_0}(s; \rho) \mathbb{E}_{a \sim \mu^*(s), b \sim \nu_0(s)} [\text{Var}_{P_{s,a,b}}(V_{\text{pe}}^-)]} \\
 &\stackrel{(iv)}{\leq} 2 \sqrt{\frac{C_{\text{clipped}}^* S(A+B) \log \frac{N}{\delta}}{N}} \sqrt{\sum_{s \in \mathcal{S}} d^{\mu^*, \nu_0}(s; \rho) \text{Var}_{P_{\mu^*, \nu_0}}(V_{\text{pe}}^-)}.
 \end{aligned}$$

reference (left) vs harness candidate (right)

1-turn fails to compile (emits LaTeX-isms like \overset as Typst).

Harness compiles and looks close — but content plateaus at 60.7 after

23 turns / 8 score runs. The residual is semantic: subscript stacks like

$\text{Var}_{\{P_{\{s,a,\nu_0(s)\}}\}}$  flatten or reorder,  $\min$  spacing differs, radical

groupings shift token positions.

Hardness = dense-notation translation; the ceiling is semantic, not layout.

Case: pubmed\_table\_004 — Clinical retention-time table (86 rows) 1-turn 27.3 -> harness 60.2

| NO. | name                      | tR (min) | Exp.  | tR MLR | tR ANN | NO. | name                        |
|-----|---------------------------|----------|-------|--------|--------|-----|-----------------------------|
| m1  | Dichlorodifluoromethane   | 1.35     |       | 1.50   | 1.52   | m44 | p-Xylene                    |
| m2  | Carbon disulfide          | 4.11     |       | 2.71   | 2.67   | m45 | m-Xylene                    |
| m3  | Allyl chloride            | 4.11     |       | 6.57   | 6.57   | m46 | o-Xylenea                   |
| m4  | Methylene chloride        | 4.40     |       | 3.93   | 4.12   | m47 | Styrene                     |
| m5  | 1,1-Dichloroethene        | 4.57     |       | 6.75   | 6.79   | m48 | Bromoforb                   |
| m6  | Acetone                   | 4.57     |       | 3.66   | 4.22   | m49 | Isopropylbenzene (Cumene)   |
| m7  | trans-1,2-Dichloroethene  | 4.57     |       | 8.77   | 8.76   | m50 | cis-1,4-Dichloro-2-butene   |
| m8  | Acrylonitrile             | 5.00     |       | 4.52   | 4.80   | m51 | Trichlorofluoromethane      |
| m9  | 1,1-Dichloroethane        | 6.14     |       | 8.62   | 8.16   | m52 | 1,1,2,2-Tetrachloroethane   |
| m10 | Vinyl acetate             | 6.43     |       | 11.47  | 11.50  | m53 | Bromobenzene                |
| m11 | 2,2-Dichloropropane       | 8.10     |       | 12.42  | 12.61  | m54 | 1,2,3-Trichloropropanea     |
| m12 | Chloromethane             | 1.49     | -0.34 | -0.12  |        | m55 | n-Propylbenzene             |
| m13 | cis-1,2-Dichloroethene    | 8.25     |       | 9.82   | 9.86   | m56 | 2-Chlorotoluene             |
| m14 | Propionitrile             | 8.51     |       | 5.61   | 5.99   | m57 | trans-1,4-Dichloro-2-butene |
| m15 | Chloroform                | 9.01     |       | 7.78   | 7.50   | m58 | 1,3,5-Trimethylbenzene      |
| m16 | Methacrylonitrile         | 9.19     |       | 8.47   | 8.74   | m59 | 4-Chlorotoluene             |
| m17 | 1,1,1-Trichloroethane     | 10.18    |       | 13.58  | 12.60  | m60 | Pentachloroethane           |
| m18 | Carbon tetrachloride      | 11.02    |       | 11.75  | 11.23  | m61 | 1,2,4-Trimethylbenzenea     |
| m19 | 1,1-Dichloropropene       | 11.50    |       | 14.4   | 14.29  | m62 | Acrolein                    |
| m20 | Benzene                   | 1.56     |       | 1.70   | 1.84   | m63 | sec-Butylbenzene            |
| m21 | Vinyl Chloride            | 12.09    |       | 11.39  | 11.53  | m64 | tert-Butylbenzene           |
| m22 | 1,2-Dichloroethane        | 14.03    |       | 13.84  | 13.67  | m65 | p-Isopropyltoluene          |
| m23 | Trichloroethene           | 14.51    |       | 15.14  | 15.43  | m66 | 1,3-Dichlorobenzene         |
| m24 | 1,2-Dichloropropane       | 15.39    |       | 16.45  | 16.45  | m67 | 1,4-Dichlorobenzene         |
| m25 | Dibromomethane            | 15.43    |       | 12.66  | 12.21  | m68 | Benzyl chloride             |
| m26 | methacrylatea Methyl      | 15.50    |       | 15.79  | 15.69  | m69 | n-Butylbenzene              |
| m27 | 1,4-Dioxane               | 16.17    |       | 15.46  | 15.43  | m70 | 1,2-Dichlorobenzene         |
| m28 | 2-Chloroethyl vinyl ether | 17.32    |       | 15.02  | 15.92  | m71 | 1,2-Dibromo-3-chloropropane |
| m29 | 4-Methyl-2-pentanone      | 17.47    |       | 16.13  | 16.81  | m72 | 1,2,4-Trichlorobenzene      |
| m30 | trans-1,3-Dichloropropene | 2.19     |       | 2.08   | 1.94   | m73 | Iodomethane                 |
| m31 | Bromomethane              | 18.29    |       | 18.63  | 18.53  | m74 | Hexachlorobutadiene         |
| m32 | Toluene                   | 19.38    |       | 15.38  | 15.90  | m75 | Naphthalene                 |
| m33 | cis-1,3-Dichloropropene   | 19.59    |       | 19.34  | 19.50  | m76 | 1,2,3-Trichlorobenzene      |
| m34 | methacrylatea Ethyl       | 20.01    |       | 24.52  | 24.70  | m77 | 1,4-Difluorobenzene         |
| m35 | 2-Hexanone                | 20.30    |       | 21.03  | 21.24  | m78 | Chlorobenzene-d5            |
| m36 | Tetrachloroethene         | 20.26    |       | 19.88  | 19.62  | m79 | 1,4-Dichlorobenzene-d4      |
| m37 | 1,3-Dichloropropanea      | 20.51    |       | 16.58  | 16.67  | m80 | 4-Bromofluorobenzene        |
| m38 | Dibromochloromethane      | 21.19    |       | 21.94  | 21.62  | m81 | 1,2-Dichlorobenzene-d4a     |
| m39 | 1,2-Dibromoethane         | 21.52    |       | 20.89  | 20.21  | m82 | Dichloroethane-d4           |
| m40 | Chloroethane              | 2.21     |       | 2.83   | 3.03   | m83 | Acetonitrile                |
| m41 | Chlorobenzene             | 23.17    |       | 23.52  | 23.61  | m84 | Toluene-d8                  |
| m42 | 1,1,1,2-Tetrachloroethane | 23.36    |       | 22.88  | 22.81  | m85 | Fluorobenzene               |
| m43 | Ethylbenzene              | 23.38    |       | 23.26  | 23.06  |     |                             |

| NO. | name                      | tR (min) | Exp.  | tR MLR | tR ANN | NO. | name                         | tR (min) | Exp.  | tR MLR | tR ANN |
|-----|---------------------------|----------|-------|--------|--------|-----|------------------------------|----------|-------|--------|--------|
| m1  | Dichlorodifluoromethane   | 1.35     |       | 1.50   | 1.52   | m44 | p-Xylene                     | 23.54    | 22.69 | 22.8   |        |
| m2  | Carbon disulfide          | 4.11     |       | 2.71   | 2.67   | m45 | m-Xylene                     | 23.54    | 22.69 | 22.6   |        |
| m3  | Allyl chloride            | 4.11     |       | 6.57   | 6.57   | m46 | o-Xylenea                    | 25.16    | 22.66 | 22.5   |        |
| m4  | Methylene chloride        | 4.40     |       | 3.93   | 4.12   | m47 | Styrene                      | 25.3     | 23.09 | 22.9   |        |
| m5  | 1,1-Dichloroethene        | 4.57     |       | 6.75   | 6.79   | m48 | Bromoforb                    | 26.23    | 26.19 | 26.2   |        |
| m6  | Acetone                   | 4.57     |       | 3.66   | 4.22   | m49 | Isopropylbenzene (Cumene)    | 26.37    | 27.00 | 26.76  |        |
| m7  | trans-1,2-Dichloroethene  | 4.57     |       | 8.77   | 8.76   | m50 | cis-1,4-Dichloro-2-butene    | 27.12    | 27.18 | 27.39  |        |
| m8  | Acrylonitrile             | 5.00     |       | 4.52   | 4.80   | m51 | Trichlorofluoromethane       | 27.42    | 6.04  | 5.75   |        |
| m9  | 1,1-Dichloroethane        | 6.14     |       | 8.62   | 8.16   | m52 | 1,1,2,2-Tetrachloroethane    | 27.29    | 25.7  | 25.79  |        |
| m10 | Vinyl acetate             | 6.43     |       | 11.47  | 11.50  | m53 | Bromobenzene                 | 27.46    | 26.85 | 26.92  |        |
| m11 | 2,2-Dichloropropane       | 8.10     |       | 12.42  | 12.61  | m54 | 1,2,3-Trichloropropanea      | 27.55    | 27.46 | 26.61  |        |
| m12 | Chloromethane             | 1.49     | -0.34 | -0.12  |        | m55 | n-Propylbenzene              | 27.58    | 27.92 | 27.64  |        |
| m13 | cis-1,2-Dichloroethene    | 8.25     |       | 9.82   | 9.86   | m56 | 2-Chlorotoluene              | 28.19    | 27.88 | 27.96  |        |
| m14 | Propionitrile             | 8.51     |       | 5.61   | 5.99   | m57 | trans-1,4-Dichloro-2-butene  | 28.26    | 26.64 | 26.75  |        |
| m15 | Chloroform                | 9.01     |       | 7.78   | 7.50   | m58 | 1,3,5-Trimethylbenzene       | 28.31    | 26.62 | 28.48  |        |
| m16 | Methacrylonitrile         | 9.19     |       | 8.47   | 8.74   | m59 | 4-Chlorotoluene              | 28.33    | 27.85 | 27.94  |        |
| m17 | 1,1,1-Trichloroethane     | 10.18    |       | 13.58  | 12.60  | m60 | Pentachloroethane            | 29.41    | 28.41 | 28.17  |        |
| m18 | Carbon tetrachloride      | 11.02    |       | 11.75  | 11.23  | m61 | 1,2,4-Trimethylbenzenea      | 29.47    | 26.75 | 26.61  |        |
| m19 | 1,1-Dichloropropene       | 11.50    |       | 14.4   | 14.29  | m62 | Acrolein                     | 3.19     | 3.33  | 3.44   |        |
| m20 | Benzene                   | 1.56     |       | 1.70   | 1.84   | m63 | sec-Butylbenzene             | 30.25    | 31.85 | 31.52  |        |
| m21 | Vinyl Chloride            | 12.09    |       | 11.39  | 11.53  | m64 | tert-Butylbenzene            | 30.59    | 30.14 | 29.86  |        |
| m22 | 1,2-Dichloroethane        | 14.03    |       | 13.84  | 13.67  | m65 | p-Isopropyltoluene           | 30.59    | 31.10 | 30.85  |        |
| m23 | Trichloroethene           | 14.51    |       | 15.14  | 15.43  | m66 | 1,3-Dichlorobenzene          | 30.56    | 31.78 | 31.94  |        |
| m24 | 1,2-Dichloropropane       | 15.39    |       | 16.45  | 16.45  | m67 | 1,4-Dichlorobenzene          | 31.22    | 32.65 | 32.88  |        |
| m25 | Dibromomethane            | 15.43    |       | 12.66  | 12.21  | m68 | Benzyl chloride              | 32.00    | 28.35 | 28.52  |        |
| m26 | methacrylatea Methyl      | 15.50    |       | 15.79  | 15.69  | m69 | n-Butylbenzene               | 32.23    | 32.68 | 32.35  |        |
| m27 | 1,4-Dioxane               | 16.17    |       | 15.46  | 15.43  | m70 | 1,2-Dichlorobenzene          | 32.31    | 32.00 | 32.17  |        |
| m28 | 2-Chloroethyl vinyl ether | 17.32    |       | 15.02  | 15.92  | m71 | 1,2-Dibromo-3-chloropropanea | 35.30    | 33.29 | 33.39  |        |
| m29 | 4-Methyl-2-pentanone      | 17.47    |       | 16.13  | 16.81  | m72 | 1,2,4-Trichlorobenzene       | 38.19    | 40.76 | 4102   |        |
| m30 | trans-1,3-Dichloropropene | 2.19     |       | 2.08   | 1.94   | m73 | Iodomethane                  | 3.56     | 6.13  | 5.83   |        |
| m31 | Bromomethane              | 18.29    |       | 18.63  | 18.53  | m74 | Hexachlorobutadiene          | 28.57    | -     | -      |        |
| m32 | Toluene                   | 19.38    |       | 15.38  | 15.90  | m75 | Naphthalene                  | 29.05    | 33.39 | 33.04  |        |
| m33 | cis-1,3-Dichloropropene   | 19.59    |       | 19.34  | 19.50  | m76 | 1,2,3-Trichlorobenzene       | 40.01    | 23.49 | 40.8   |        |
| m34 | methacrylatea Ethyl       | 20.01    |       | 24.52  | 24.70  | m77 | 1,4-Difluorobenzene          | 13.26    | 13.68 | 13.6   |        |
| m35 | 2-Hexanone                | 20.30    |       | 21.03  | 21.24  | m78 | Chlorobenzene-d5             | 23.10    | 23.49 | 23.28  |        |
| m36 | Tetrachloroethene         | 20.26    |       | 19.88  | 19.62  | m79 | 1,4-Dichlorobenzene-d4       | 31.16    | 32.65 | 32.88  |        |
| m37 | 1,3-Dichloropropanea      | 20.51    |       | 16.58  | 16.67  | m80 | 4-Bromofluorobenzene         | 27.83    | 25.93 | 26.08  |        |
| m38 | Dibromochloromethane      | 21.19    |       | 21.94  | 21.62  | m81 | 1,2-Dichlorobenzene-d4a      | 32.30    | 32.00 | 32.11  |        |
| m39 | 1,2-Dibromoethane         | 21.52    |       | 20.89  | 20.21  | m82 | Dichloroethane-d4            | 12.08    | 8.62  | 8.76   |        |
| m40 | Chloroethane              | 2.21     |       | 2.83   | 3.03   | m83 | Acetonitrile                 | 4.11     | 3.23  | 4.00   |        |
| m41 | Chlorobenzene             | 23.17    |       | 23.52  | 23.61  | m84 | Toluene-d8                   | 18.27    | 18.63 | 18.53  |        |
| m42 | 1,1,1,2-Tetrachloroethane | 23.36    |       | 22.88  | 22.81  | m85 | Fluorobenzene                | 13.00    | 14.04 | 14.11  |        |
| m43 | Ethylbenzene              | 23.38    |       | 23.26  | 23.06  |     |                              |          |       |        |        |

reference (left) vs harness candidate (right)

Reference renders one wide 6-column table; candidate reflows it into narrower stacked fragments with smaller type and different column breaks. All cells present (content 77.8) but the table geometry is structurally different, so layout/raster stay low. Same reflow failure that dominates the current hard set — at 5x the cell count, with multi-line cells and paired half-tables.

## I. INTRODUCTION

In Ref. [1] two different expressions of quantum dual total correlation were obtained: one in terms of von Neumann entropy and other in terms of quantum relative entropy. It was claimed that the two expressions are equivalent. In a comment [2] on Ref. [1], Lee *et al.* have shown that the quantum dual total correlation of an  $n$ -partite quantum state cannot be represented as the quantum relative entropy between  $(n-1)$  copies of the quantum state and the product of  $n$  different reduced quantum states for  $n \geq 3$ . They arrived at this conclusion by considering explicitly the “support” condition of quantum relative entropy. Essentially, what Lee *et al.* have shown is that the following two expressions are not equal for  $n \geq 3$ :

$$I_n(\rho) := \sum_{k=1}^n S(\rho_k) - (n-1)S(\rho), \quad (1)$$

where  $\rho_k = \text{tr}_k(\rho)$  denotes the  $(n-1)$ -partite quantum state obtained by taking the partial trace on the  $k^{\text{th}}$  party of  $\rho$ , and

$$J_n(\rho) := S(\rho^{\otimes(n-1)} \parallel \otimes_{k=1}^n \rho_k), \quad (2)$$

where the quantum relative entropy is

$$S(\tau \parallel \sigma) := \begin{cases} \text{tr}(\tau \log \tau) - \text{tr}(\tau \log \sigma) & \text{if } \text{supp}(\tau) \subseteq \text{supp}(\sigma) \\ \infty & \text{otherwise.} \end{cases}$$

To justify their claim, authors provide two examples which imply that the above two expressions of  $n$ -partite quantum mutual information are not equivalent.  $I_n(\rho)$  in Eq. (1) is non-negative and non-increasing under local CPTP maps [3], and therefore is a suitable monotonic measure of multi-partite correlations, while  $J_n(\rho)$  in Eq. (2) is not.

II. REAFFIRMING CLAIM OF LEE *ET AL.*

The claim of Lee *et al.* is right. In this article we show analytically why the above two expressions are not equivalent. We begin with expression of  $I_n(\rho)$  [Eq. (1)] and proceed to show that this is not equal to  $J_n(\rho)$  [Eq. (2)], as argued below.

## Introduction

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reference (left) vs harness candidate (right)

1-turn: no compile. Harness: content 94.5 — it reads like the paper, but doesn't look like it: REVTeX centered small-caps headings vs Typst default bold left headings, tighter leading, different display sizing. Density mismatch compounds page by page: candidate fits in 4 pages instead of 5 (pages 5/4). Hardness = document-class-level style emulation, beyond content.

## ViewPoint: Panoramic Video Generation with Pretrained Diffusion Models

Zixun Fang<sup>1,2\*</sup> Kai Zhu<sup>1,2</sup> Zhilong Liu<sup>1</sup> Yu Liu<sup>2</sup> Wei Zhai<sup>1†</sup>  
Yang Cao<sup>1</sup> Zheng-Jun Zhu<sup>1</sup>  
<sup>1</sup>USTC <sup>2</sup>TongYi Lab <sup>3</sup>HKU

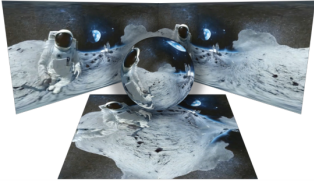


Figure 1: **The generated result.** The image at the bottom displays the ViewPoint map generated by our model, with the background image showing the concatenated equirectangular map derived from the ViewPoint map. The generated panoramic image exhibits excellent spatial consistency, as the equirectangular map can be seamlessly stitched together. Project page: [ViewPoint](#).

### Abstract

Panoramic video generation aims to synthesize 360-degree immersive videos, holding significant importance in the fields of VR, world models, and spatial intelligence. Existing works fail to synthesize high-quality panoramic videos due to the inherent modality gap between panoramic data and perspective data, which constitutes the majority of the training data for modern diffusion models. In this paper, we propose a novel framework utilizing pretrained perspective video models for generating panoramic videos. Specifically, we design a novel panorama representation named ViewPoint map, which possesses global spatial continuity and fine-grained visual details simultaneously. With our proposed Pano-Prospective attention mechanism, the model benefits from pretrained perspective priors and captures the panoramic spatial correlations of the ViewPoint map effectively. Extensive experiments demonstrate that our method can synthesize highly dynamic and spatially consistent panoramic videos, achieving state-of-the-art performance and surpassing previous methods.

\*Intern at Tongyi Lab

Preprint. Under review.

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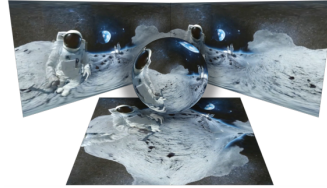


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### Introduction

Imagine you're traveling—would you wise the chance to capture the breathtaking landscapes that unfold before you? When you return home, do you long to relive and immerse yourself

\*Intern at Tongyi Lab

### 3.3 Pano-Prospective Attention

ViewPoint Map offers global panorama information and effectively utilizes the inherent in-context generation capabilities [11] of diffusion models. To further accelerate convergence and exploit the priors of the base model, we propose a Pano-Prospective Attention mechanism. Specifically, Pano-Attention is responsible for contextual learning to maintain spatial consistency, while Prospective-Attention focuses on generating fine-grained visual information for each subregion. As shown in the second row of Fig. 2, we first concatenate the noisy Viewpoint Map, the projected input perspective view frames, and the corresponding masks along the channel dimension. In the latent space, the shape of the concatenated features is  $(batch\_size, channels, frames, height, width)$ . After passing through the Pano-Attention block, we reshape it to  $(4 \times batch\_size, channels, frames, height/2, width/2)$  and feed it into the Prospective-Attention for per-regional modeling. Each attention block consists of a self-attention layer, a cross-attention layer, and an FFN, the input prompt is integrated through the cross-attention mechanism.

### 3.4 Overlapping Fusion

Each subregion in a ViewPoint map has a partial spatial overlap with the two adjacent subregions. To achieve smoother transitions between subregions, we propose a gradient fusion mechanism. As shown in Fig. 3, the overlapping parts between two subregions form a regular rhombus, therefore we construct an overlapping fusion weight  $W \in \mathbb{R}^{s \times s}$  where the values decay from the area near the central face to the edges.

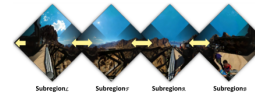


Figure 3: **Overlapping fusion.** The four subregions partially overlap with each other, thus we propose a gradient fusion mechanism to interpolate the overlapping areas, thereby enhancing spatial consistency.

For each subregion  $S_{ij} \in \mathbb{R}^{s \times s}$ ,  $s = 2n$ ,  $d \in \{C, F, R, B\}$ , the overlapping fusion process can be described by the following formula:

$$W_{ij} = \frac{1+j-2}{2(n-1)}, \quad i, j = 1, 2, \dots, n \quad (5)$$

where  $i, j$  denote the row and column indices, respectively. Next, we define rotation. For a given matrix  $A$ , we have

$$R_{90}(A) = A^T, \quad (6)$$

$$R_{180}(A) = J \cdot A^T, \quad (7)$$

$$R_{270}(A) = J \cdot A, \quad (8)$$

where  $J$  is the exchange matrix. Finally, for each subregion, we have

$$S_{i,j-1} = \begin{bmatrix} S_{i,j-1} & R_{90}(W) + R_{180}(S_{i,j-1-n}) & R_{180}(W), (i,j) \in [1,n] \times [n+1,n] \\ S_{n,j-1} & R_{90}(W) + R_{180}(S_{n,j-1-n}) & R_{180}(W), (i,j) \in [n+1,n] \times [1,n] \end{bmatrix} \quad (9)$$

$$S_{n,j} = \begin{bmatrix} S_{n,j} & R_{90}(W) + R_{180}(S_{n,j-1-n}) & R_{180}(W), (i,j) \in [1,n] \times [n+1,n] \\ S_{i,j} & R_{90}(W) + R_{180}(S_{i,j-1-n}) & R_{180}(W), (i,j) \in [n+1,n] \times [1,n] \end{bmatrix} \quad (10)$$

$$S_{i,j} = \begin{bmatrix} S_{i,j} & W + R_{90}(S_{i,j-1-n}) & R_{180}(W), (i,j) \in [1,n] \times [1,n] \\ S_{i,j-1} & W + R_{90}(S_{i,j-1-n}) & R_{180}(W), (i,j) \in [n+1,n] \times [n+1,n] \end{bmatrix} \quad (11)$$

5

exploit the priors of the base model, we propose a Pano-Prospective Attention mechanism. Specifically, Pano-Attention is responsible for contextual learning to maintain spatial consistency, while Prospective-Attention focuses on generating fine-grained visual information for each subregion. As shown in the second row of Fig. 2, we first concatenate the noisy Viewpoint Map, the projected input perspective view frames, and the corresponding masks along the channel dimension. In the latent space, the shape of the concatenated features is  $(batch\_size, channels, frames, height, width)$ . After passing through the Pano-Attention block, we reshape it to  $(batch\_size, channels, frames, height/2, width/2)$  and feed it into the Prospective-Attention, so that the model can achieve two functions: (1) generate spatial-temporal consistent panoramic videos by Pano-Attention; (2) achieve high-quality panoramic content within each subregion via Prospective-Attention.

### Overlapping Fusion

Each subregion in a ViewPoint map has a partial spatial overlap with the two adjacent subregions. To achieve smoother transitions between subregions, we propose a gradient fusion mechanism. As shown in Fig. 3, the overlapping parts between two subregions form a regular rhombus, therefore we construct an overlapping fusion weight  $W \in \mathbb{R}^{s \times s}$  where the values decay from the area near the central face to the edges.

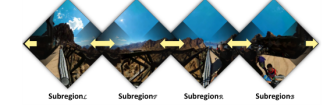


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$$W_{ij} = \frac{1+j-s}{2s-2i-2}, \quad i, j = 1, 2, \dots, s$$

where  $i, j$  denote the row and column indices, respectively. Next, we define notation: For a given matrix  $A$ , we have

$$R_0(A) = A^T$$

$$R_{90}(A) = A \cdot J$$

where  $J$  is the exchange matrix. Finally, for each subregion, we have

$$S_{i,j-1} = \begin{bmatrix} S_{i,j-1} & R_{90}(W) + R_{180}(S_{i,j-1-n}) & R_{180}(W), (i,j) \in [1,n] \times [n+1,n] \\ S_{n,j-1} & R_{90}(W) + R_{180}(S_{n,j-1-n}) & R_{180}(W), (i,j) \in [n+1,n] \times [1,n] \end{bmatrix}$$

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reference (left) vs harness candidate (right)

The largest run: \$2.16, 12.3 min, 24 turns. Harness holds page count on an 11-page document (11/11, pagination 96.3) and content 83.8. Layout stalls at 51.9 — the same class-style drift as the arXiv paper, compounded across sections, figures and two-column-ish blocks. Verbose-hard: pagination is winnable, faithful styling is the ceiling.